Supplementary Information for Four-Corner Spectral Tuning for Search-Free ADMM in Diffeomorphic Image Registration

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S1. Reproducibility and implementation

All methods use the same three-level image pyramid, phase-correlation initialization, linearization, stopping tolerances, interpolation, line search, and discrete-Jacobian safeguard. CPU timings use one BLAS/OMP thread per registration. The repository contains the exact environment, seeds, command lines, raw pair-level outputs, and scripts that regenerate all tables and figures. FIRE preprocessing uses a contrast-normalized green channel; supplied landmarks are used only for evaluation. The implementation composes displacement increments directly and enforces local orientation preservation by requiring a positive discrete-Jacobian floor during line search and pyramid transfer. This numerical safeguard does not constitute a global diffeomorphism theorem.

S2. Baseline implementations

The externally validation-selected fixed pair $(\rho, \alpha) = (0.1, 1)$ was selected on the controlled public-image pilot, not on FIRE. Residual balancing starts at $\rho = 1$ and updates the penalty from primal/dual residual imbalance with dual rescaling. The BB method is a safeguarded Barzilai–Borwein penalty proxy with clipping and periodic updates. Fixed oADMM uses $(1, 1.8)$. These details, including update intervals and safeguards, are implemented in `src/registration2d.py`.

S3. Legacy and signed rate analyses

The main text establishes basic contraction for every admissible pair when $H+G \succ 0$. The following analyses are therefore optional quantitative rate bounds, not stability tests: the global perturbation bound, block-Gershgorin disks, commutator rectangles, Perron comparison, and truncated-kernel bounds. The norm-based global perturbation objective is retained as a negative-control diagnostic: its minimization was over-conservative in variable-coefficient tests. Detailed derivations and the corresponding scripts are retained in the repository (`src/structured_envelopes.py` and `experiments/`).

S4. Additional spectral experiments

The 10,000 randomized curvature configurations confirm equality between pixelwise curvature enumeration and the four-corner reduction. In periodic constant-coefficient systems the modal envelope matches the full spectral radius to numerical precision. In smooth variable-coefficient cases the median spectral-radius gap to the numerical oracle is 0.027. Signed enclosures showed no observed violations but were conservative; certificate scaling at 128^2 remained unsuitable for online use.

S5. FIRE subgroup and parameter summaries

The audited FIRE table contains 134 pairs and five methods per pair. Percentage reductions are computed pairwise before aggregation. For the one-shot predictor, median $\rho_p = 0.1008$ (IQR 0.0397; range 0.0688–0.1402) and median $\alpha_p = 1.5699$ (IQR 0.1287; range 1.4483–1.6883). The median h_+ is 0.2249. Machine-readable summaries and A/P/S subgroup statistics accompany the repository results.

S6. One-shot reuse ablation

On a balanced 18-pair FIRE subset, per-level retuning reduced median inner iterations from 243 to 222 but increased median tuning time from 0.030 to 0.110 seconds and median total time from 2.651 to 3.895 seconds. One-shot reuse is therefore used in the main experiments.

S7. Data availability

FIRE and CIMA are public datasets subject to their respective licenses. Raw images are not redistributed. The repository stores preparation scripts, integrity checks, processed-result schemas, and all nonrestricted pair-level measurements.